

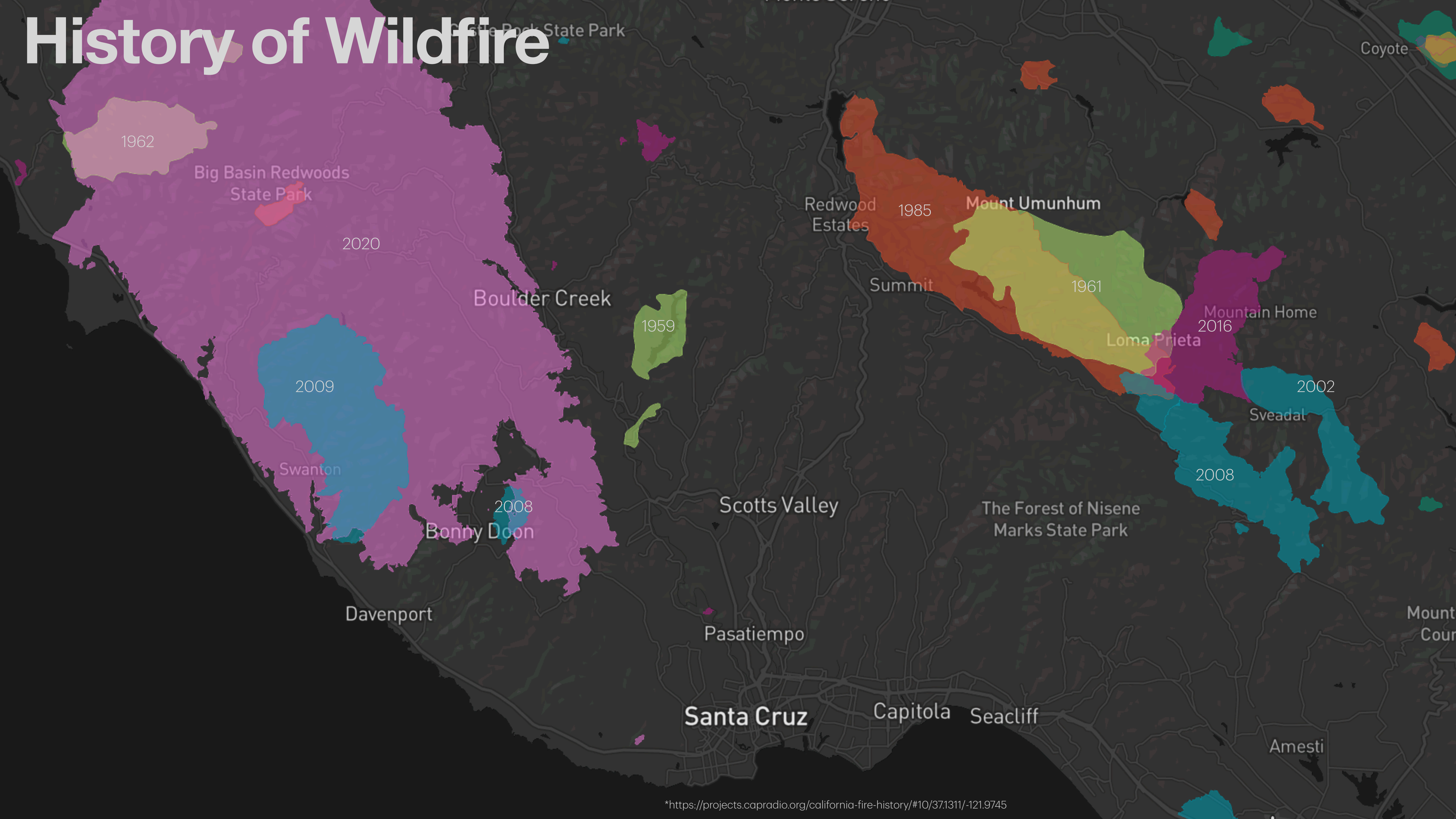
Evacuation Modeling

Firewise Communities

Along Summit Corridor East of Highway 17

Amy Winkler, Firewise Leader | 2026 September 13

History of Wildfire



History of Wildfire

Slide Credit:



MANU KOENIG
1ST DISTRICT SUPERVISOR



PARADISE

85 deaths; 153,336 acres

Heavy traffic + rapidly deteriorating fire conditions produced widespread gridlock. NIST documented at least 19 burnovers.

Roads can become traps.



LAHAINA

102 deaths; 2,170 acres

Rapid fire, infrastructure failures and constrained evacuation left residents with little time. Older adults were disproportionately vulnerable.

Minutes matter.



CZU

1 death; 86,509 acres

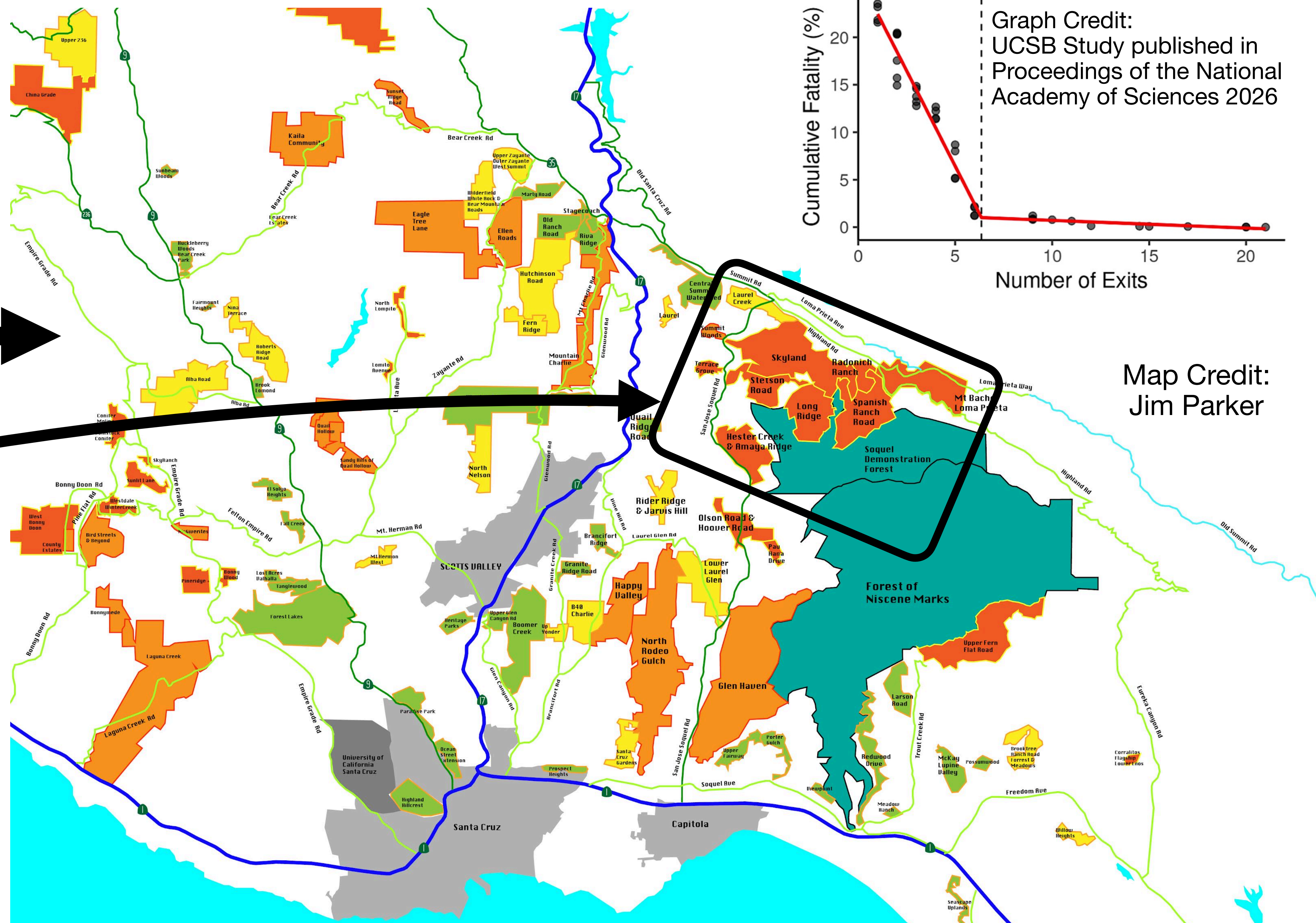
1,490 structures destroyed and one civilian fatality. Santa Cruz experienced a mass wildfire evacuation — but with more time than Paradise or Lahaina.

It can happen here.

Evacuation Vulnerability

Firewise Communities color coded by estimated evacuation risk

Our Evacuation Model focuses on this area



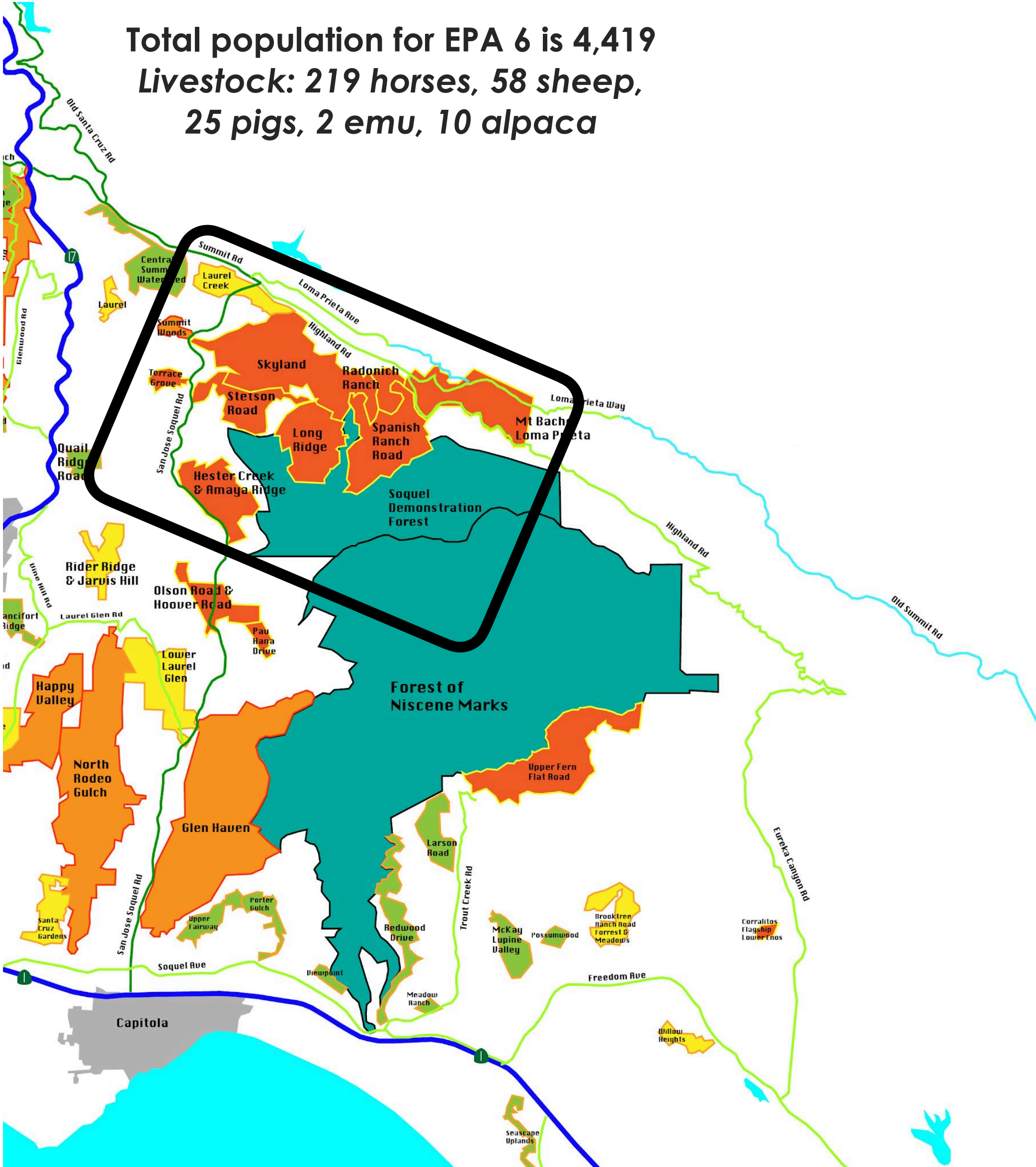
Evacuation Planning Area 6 - Summit

“**Potential evacuation options are limited.** Residents who have access to SR 17 should be directed to evacuate either north toward San Jose or south to Santa Cruz, depending upon conditions. Persons residing east of SR 17 could potentially take Soquel San Jose Road south to Santa Cruz.

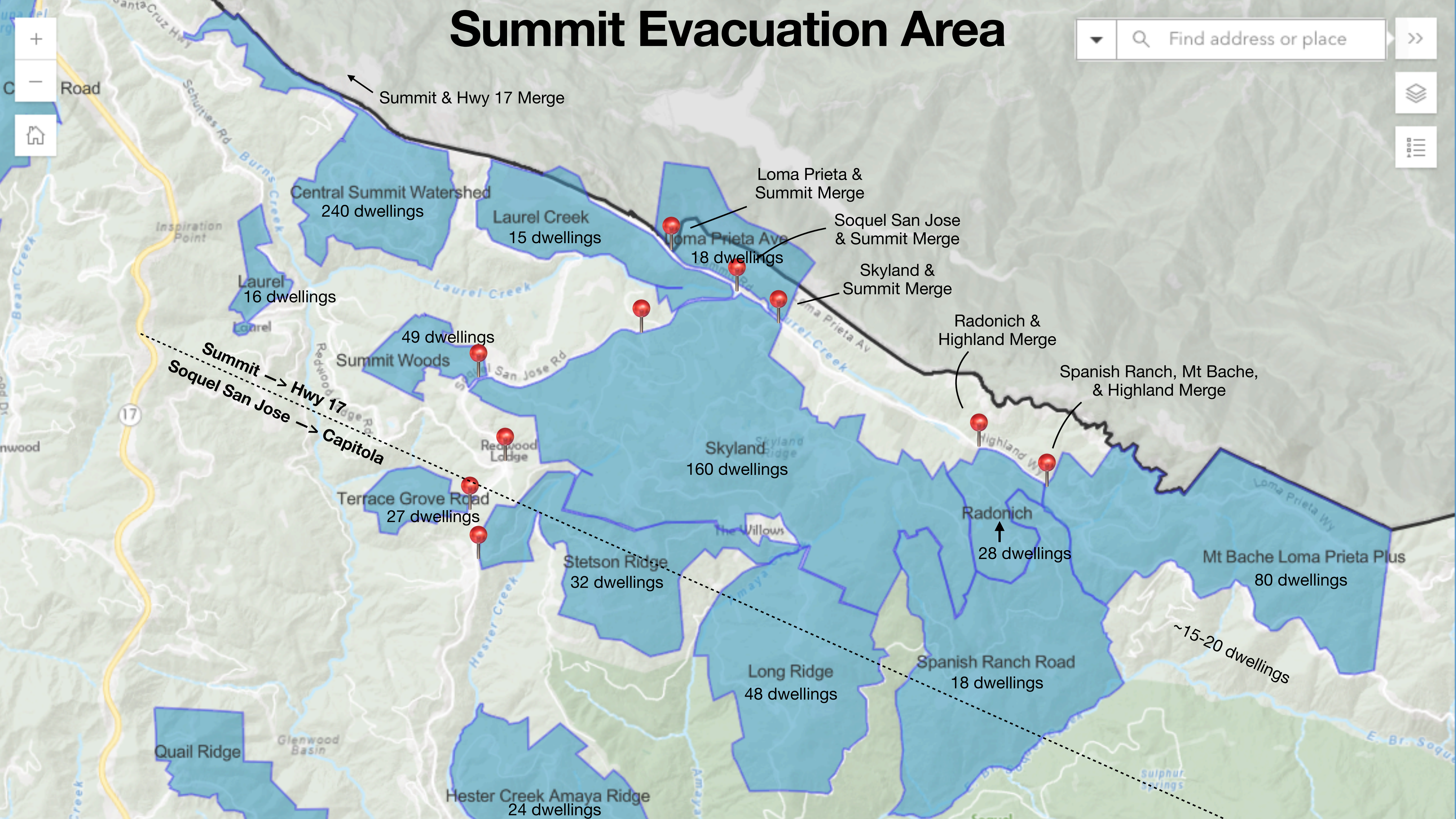
- Santa Cruz County Emergency Operations Plan

Firewise Survey Results

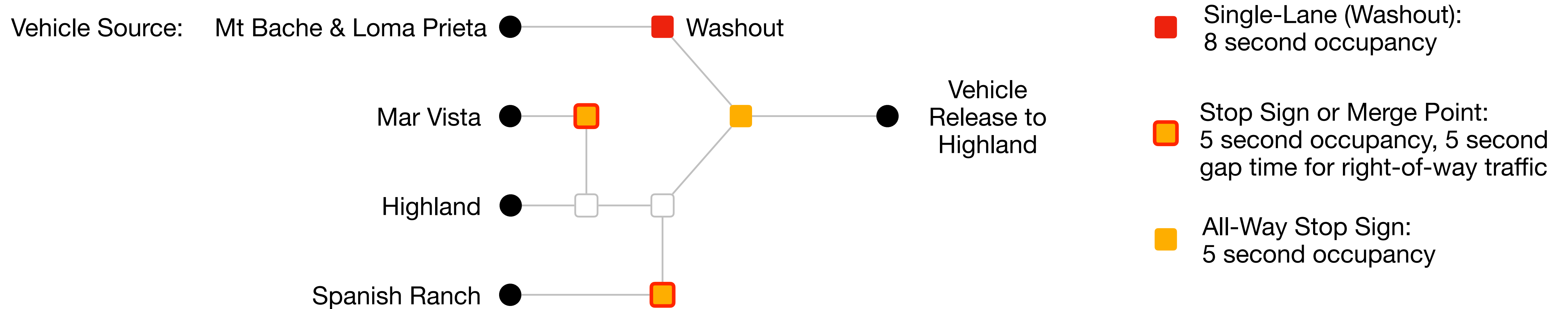
Metric	Value
Survey responses	18
Firewise Communities represented	6
Households / Residents in Firewise	226 / 562
Average vehicles per household	2
Average trailers & RVs per household	0.3
Ready to leave within 1 hour	90% of respondents
HAM/MURS operators	5
Buddy networks	9
Interested volunteers	7



Summit Evacuation Area



The Traffic Model to Mt Bache & Highland

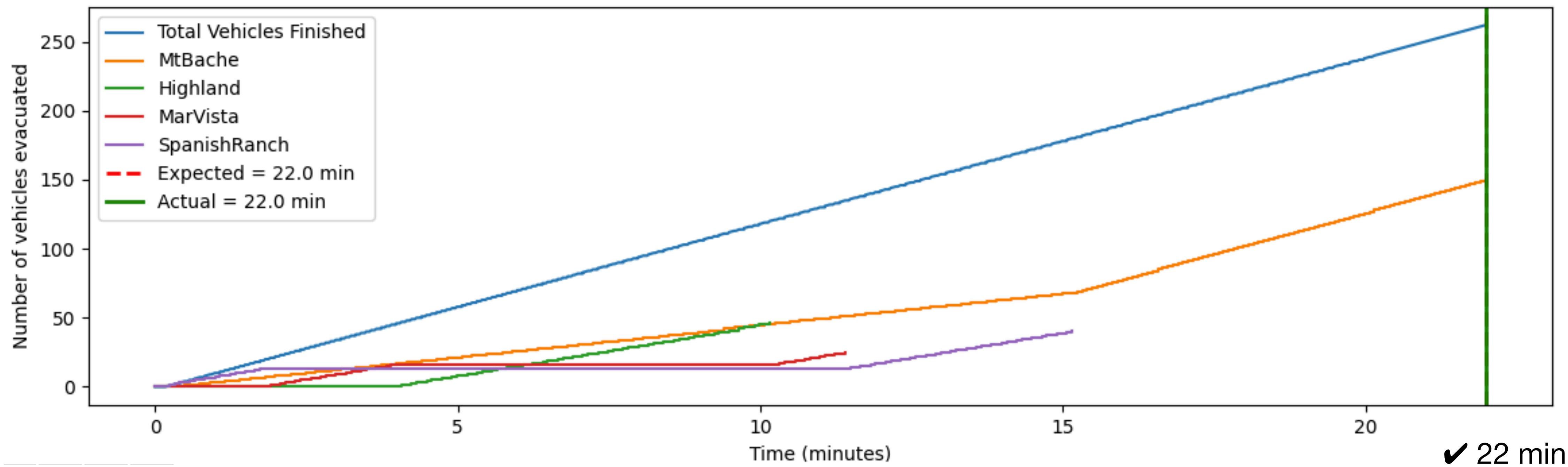


Source	Dwellings	Vehicles	MoveTo	Capacity (veh/hr)	Time (min)	MoveTo	Capacity (veh/hr)	Time (min)	MoveTo	Capacity (veh/hr)	Time (min)	Max Time (min)
Mt Bache + Loma Prieta	65	149.5	Mt Bache Washout	450	20	-	-	-	Mt Bache / Highland Intersection	720	22	22
Mar Vista	11	25.3	Mar Vista to Highland Merge	720	6	Spanish Ranch to Highland Merge	720	9				
Highland	20	46										
Spanish Ranch	18	41.4	-	-	-							

Traffic Model | Estimate vs Simulation

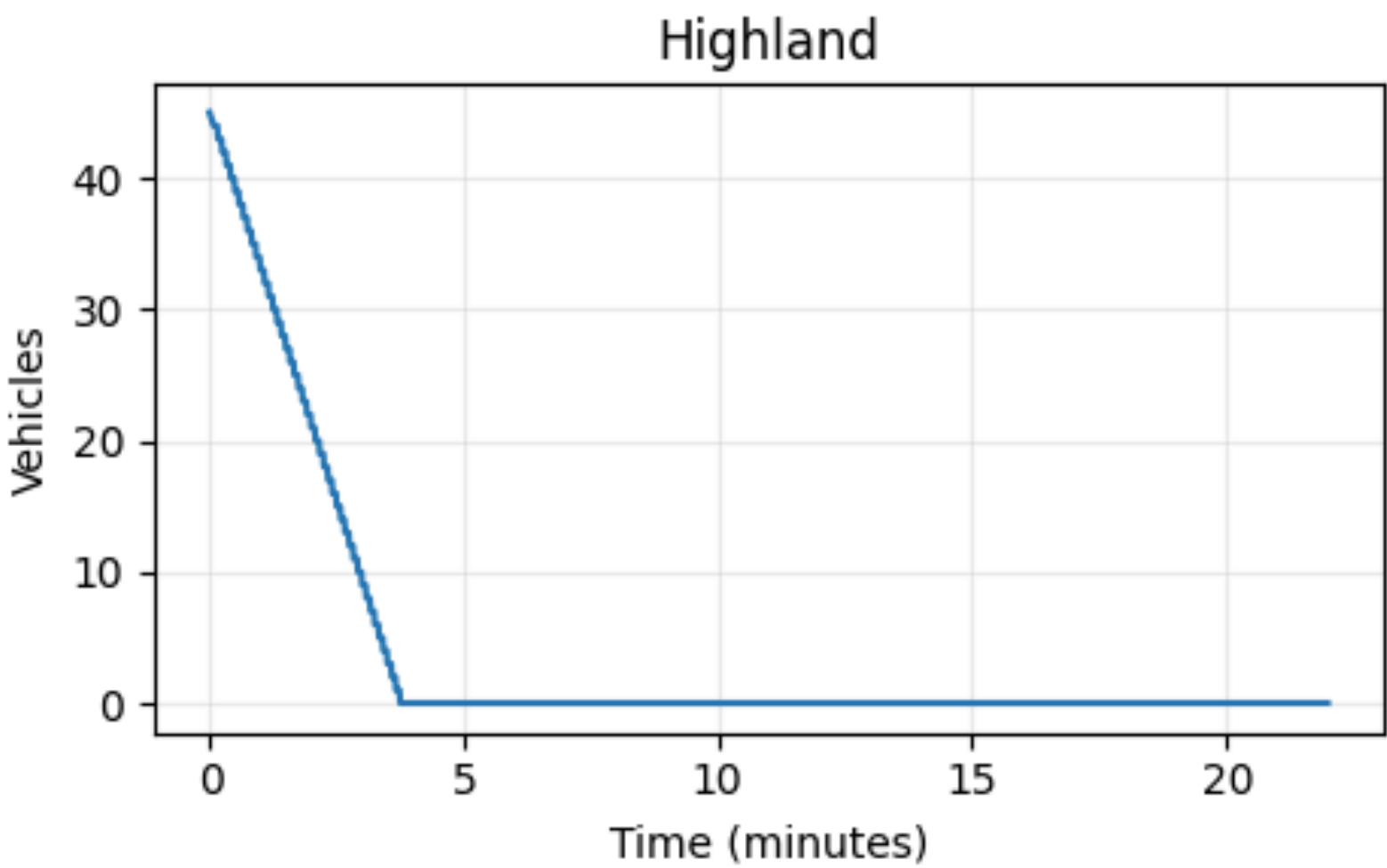
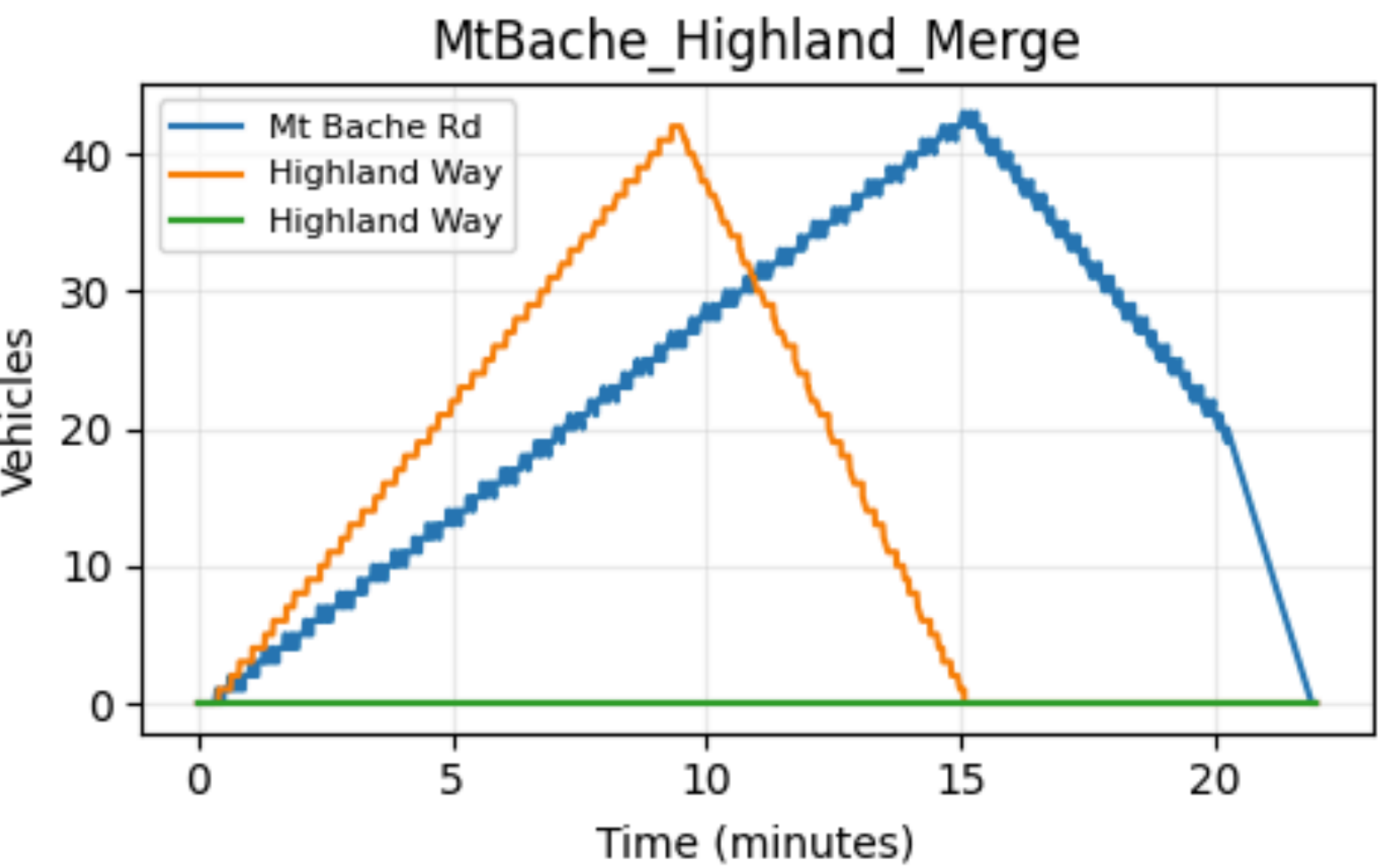
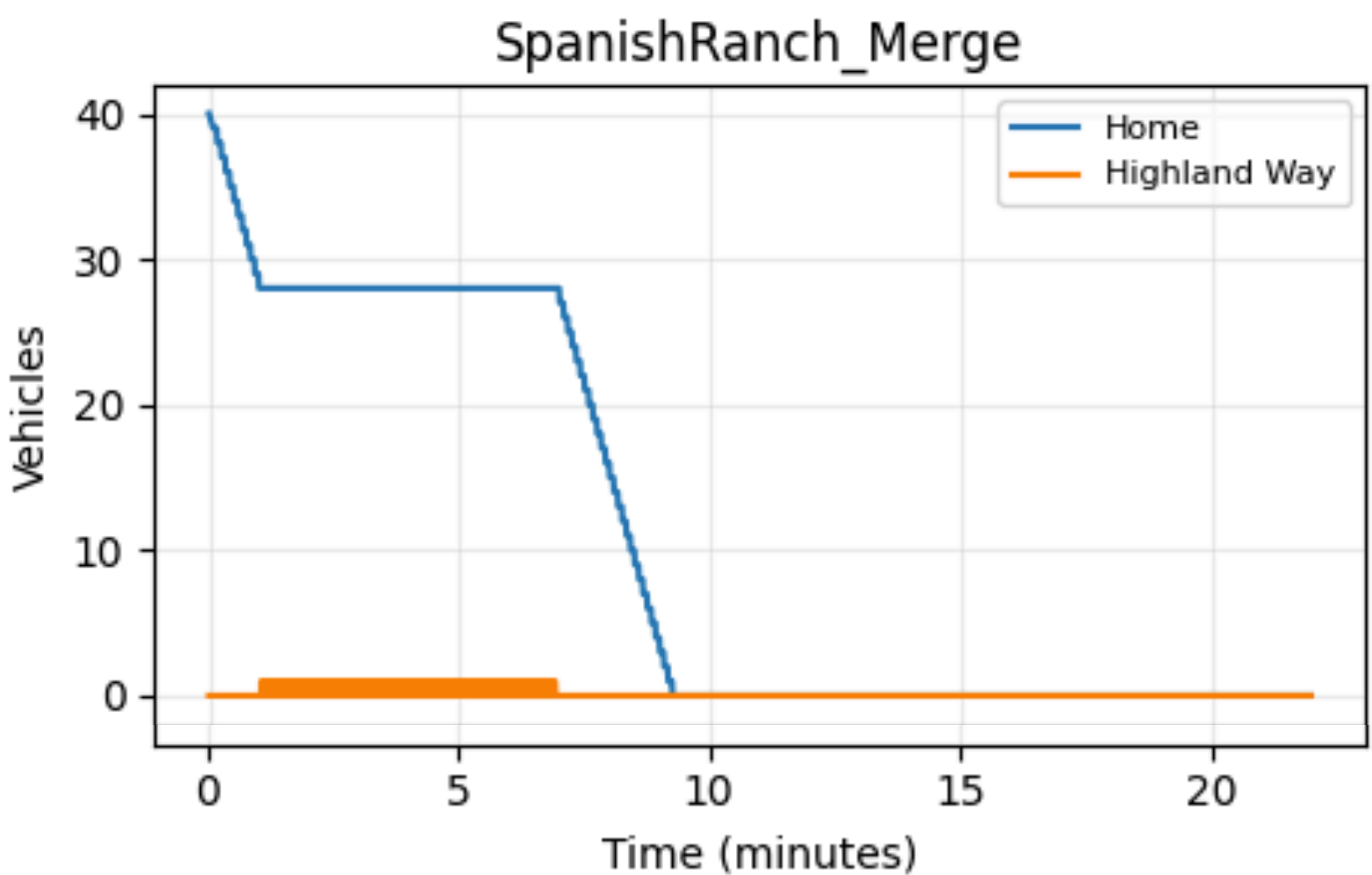
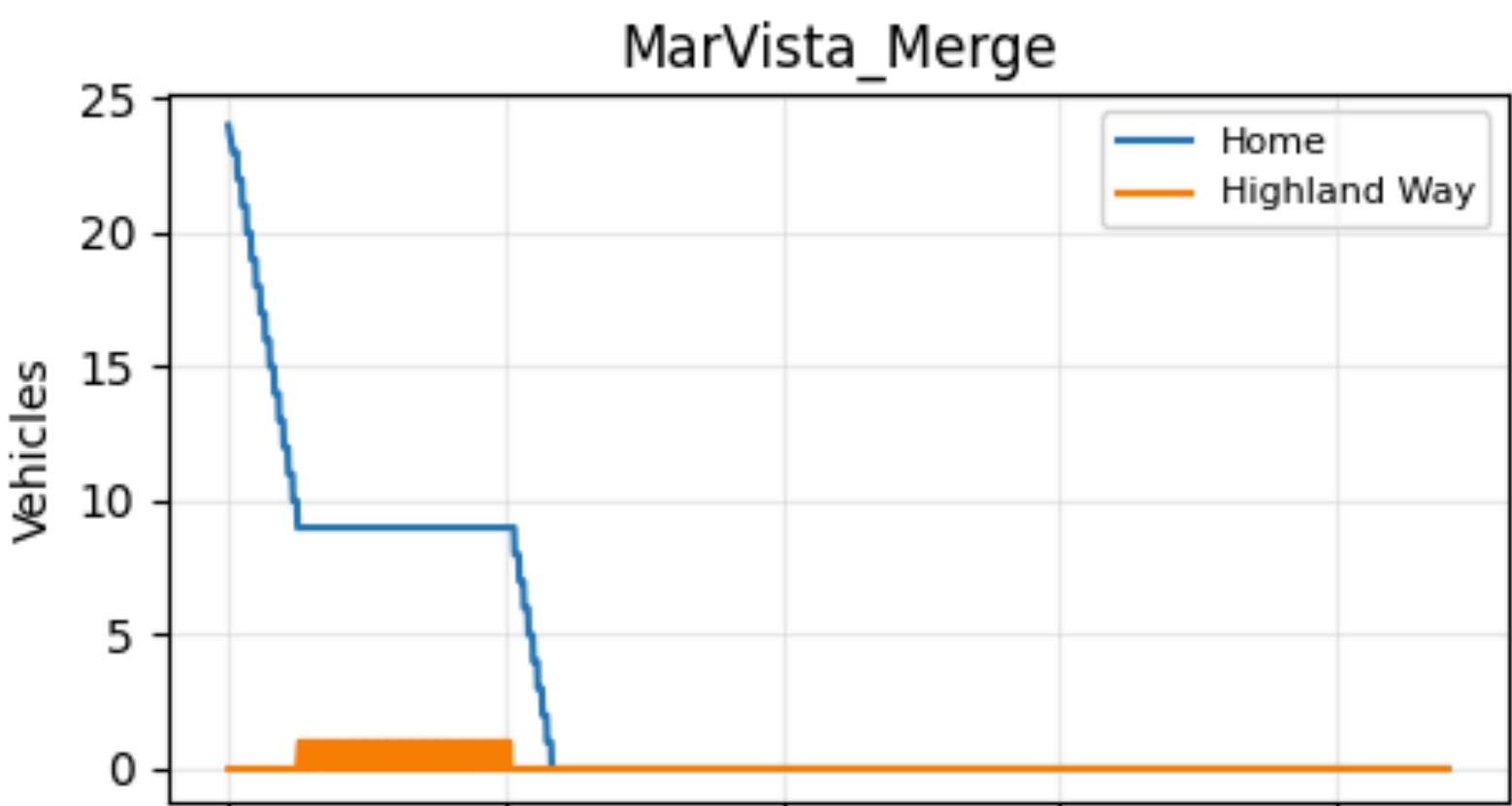
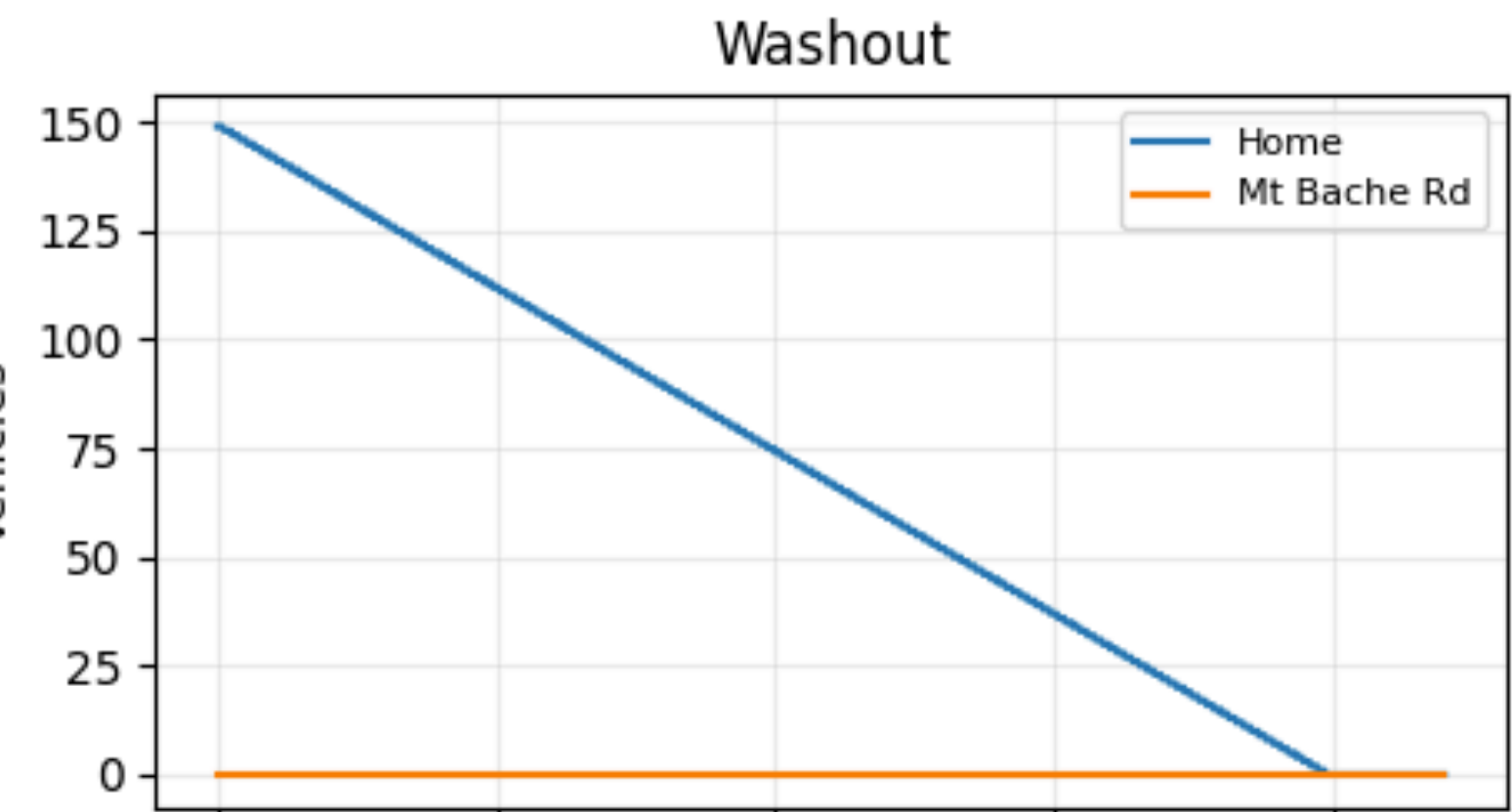
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Evacuation vs Time

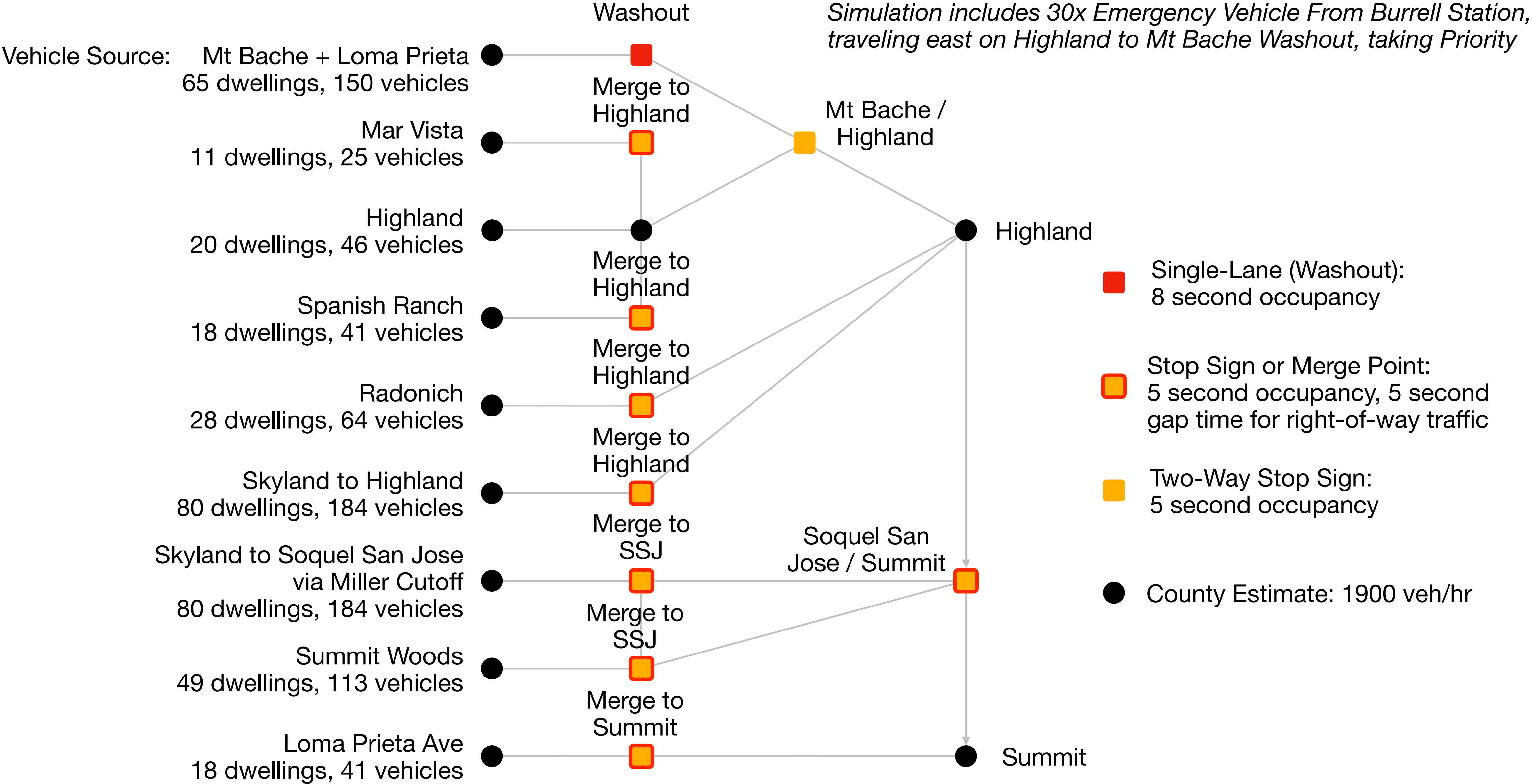


Traffic Model | Bottlenecks

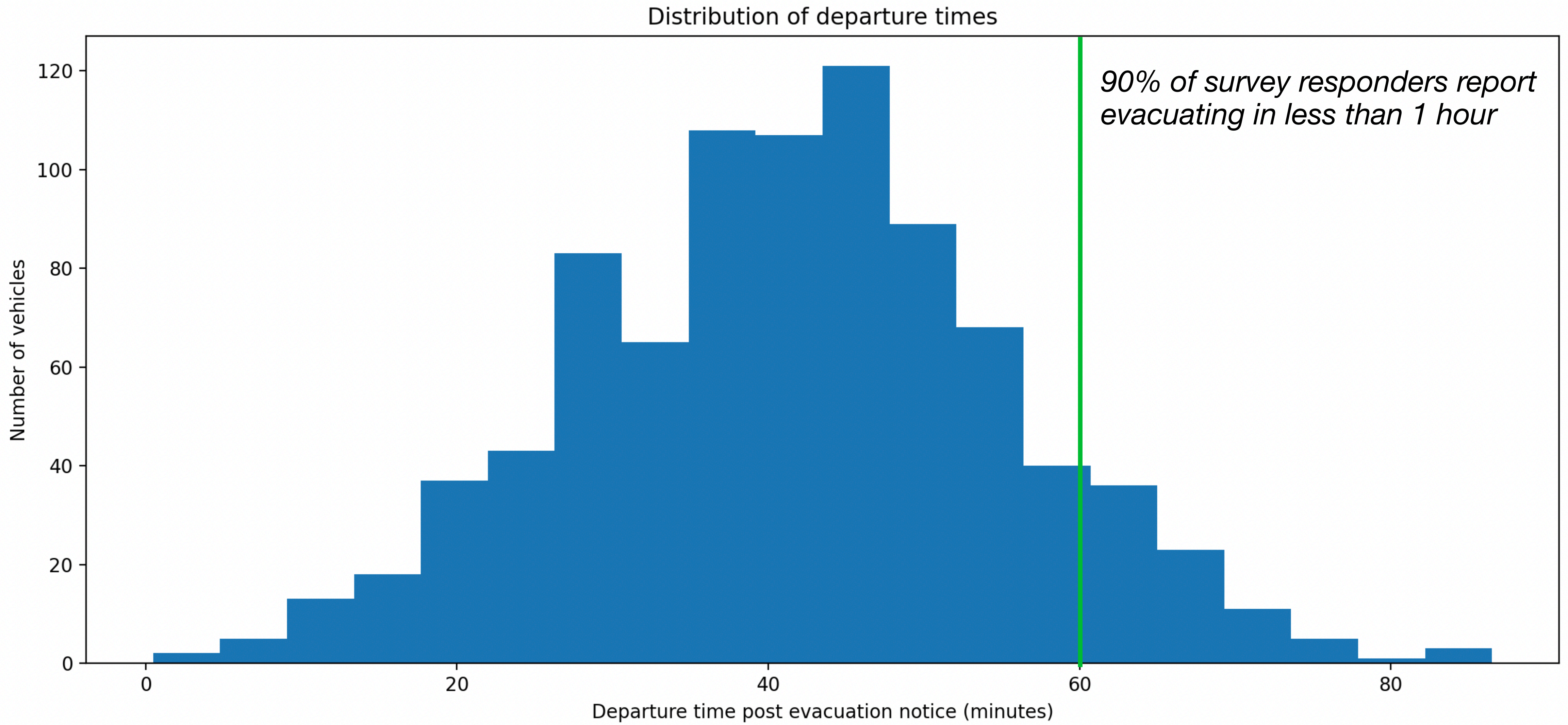
Queue Lengths Throughout Evacuation



Extending Traffic Model to Summit Area

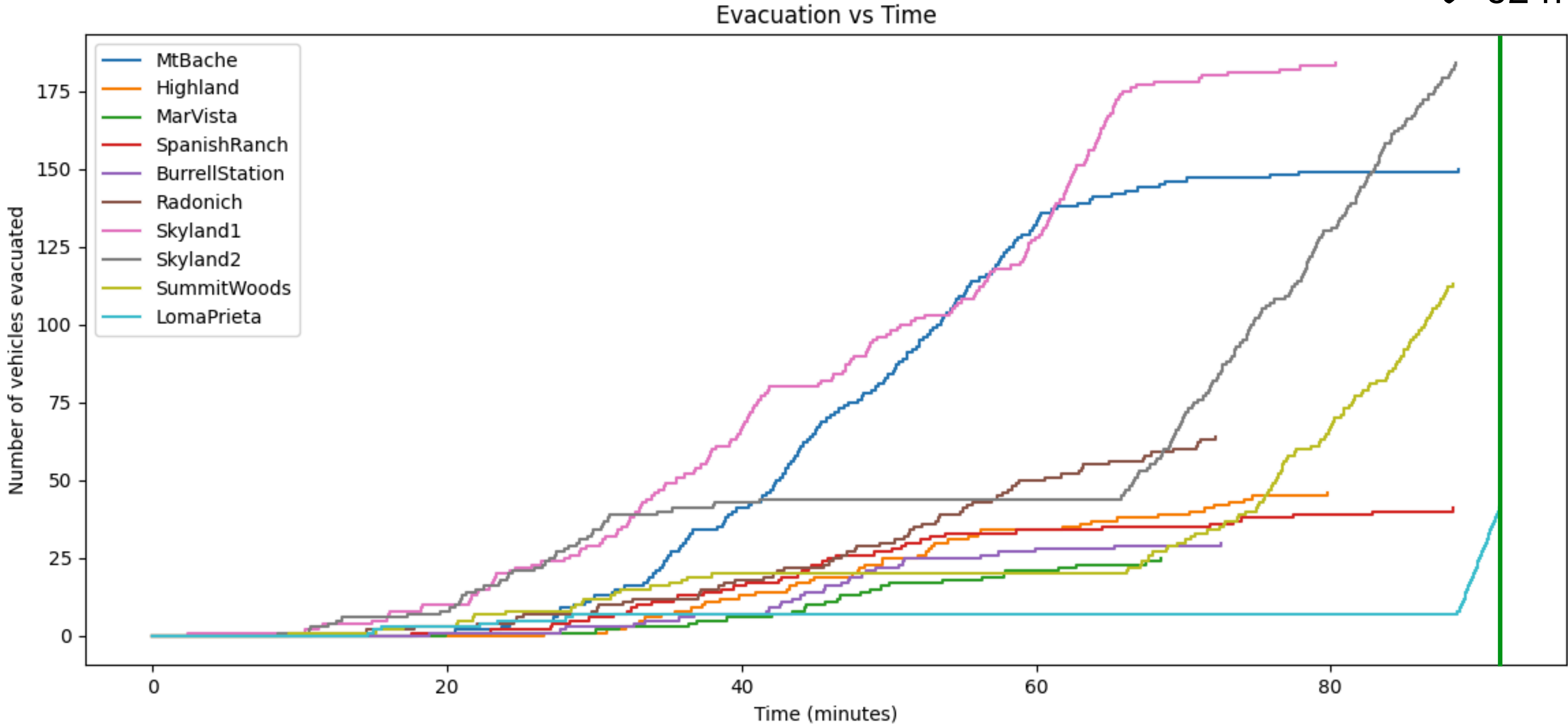


Consider Statistical Departure Time

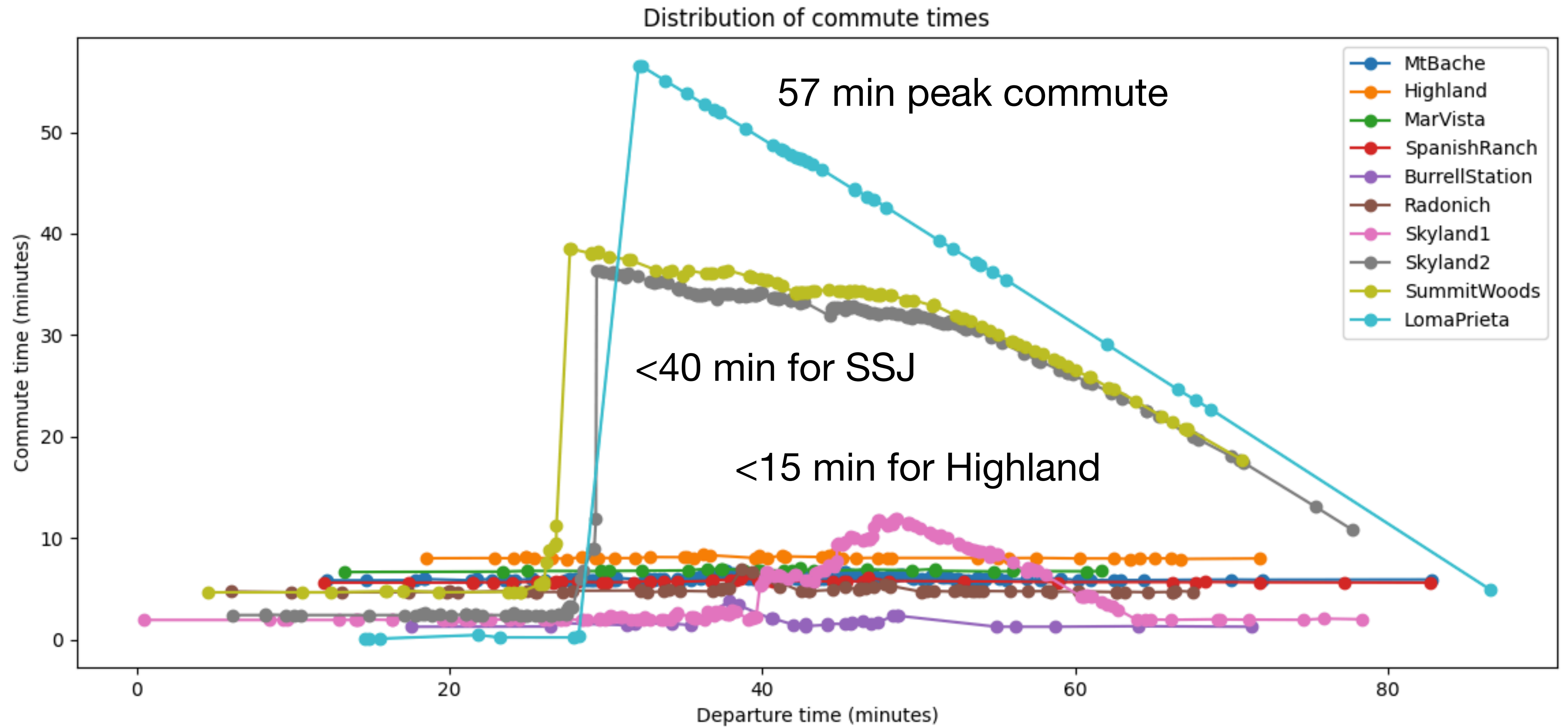


Summit Area Evacuation Time

✓ 92 min



Summit Area Evacuation Commute



Summit Area Bottlenecks | Queues > 10x

